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| **Fiscal Year:** | FY 2020 | **Task Last Updated:** | FY 05/03/2020 |
| **PI Name:** | Ju, Yiguang Ph.D. | | |
| **Project Title:** | Quantitative Studies of Cool Flame Transitions at Radiation/Stretch Extinction Using Counterflow Flames | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | COMBUSTION SCIENCE--Combustion science | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | yju@princeton.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 609-258-5644 |
| **Organization Name:** | Princeton University | | |
| **PI Address 1:** | D329B, E-Quad, Olden Street | | |
| **PI Address 2:** | | | |
| **PI Web Page:** | | | |
| **City:** | Princeton | **State:** | NJ |
| **Zip Code:** | 08540-5263 | **Congressional District:** | 12 |
| **Comments:** | | | |
| **Project Type:** | Ground,Physical Sciences Informatics (PSI) | **Solicitation / Funding Source:** | 2015 Physical Sciences NNH15ZTT001N-15PSI-A: Use of the NASA Physical Sciences Informatics System – Appendix A |
| **Start Date:** | 08/01/2016 | **End Date:** | 07/31/2021 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | Dietrich, Daniel L | **Contact Phone:** | 216-433-8759 |
| **Contact Email:** | Daniel.L.Dietrich@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 7/31/2021 per NSSC information (Ed., 9/10/20)<br>NOTE: End date changed to 7/31/2020 per NSSC information (Ed., 11/12/19)<br>NOTE: Extended to 7/31/2019 per NSSC information (Ed., 5/7/19) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Reuter, Christopher M.S. ( Princeton University ) | | |
| **Grant/Contract No.:** | NNX16AK07G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Cool flames have significant implications for the control of engine knocking, ground and space fire safety, and combustion emissions. The recent International Space Station (ISS) combustion science experiments on flame extinguishment discovered the existence of cool flames for both large single droplets and droplet arrays after radiative extinction. Numerical simulations of droplet combustion also provide an explanation for the transition of hot flame to cool flame after radiative extinction. However, it remains unclear at what conditions the hot flames transfer to cool flames and how the cool flame flammable region is related to hot flame flammability limit and extinction limit. It is also not clear how oxygen concentration, dilution, pressure, diffusion, and convection transport processes affect the cool flame transition. More importantly, due to limits of space and payload, the ISS experiments did not provide detailed speciation and temperature distributions of cool flame structures for quantitative validation of kinetics and radiation model.<br>By using the Flame Extinguishment (FLEX) experimental data from the Physical Sciences Informatics (PSI) system of combustion science, we propose to conduct new simulations and ground-based experiments using ozone-sensitized counterflow flames to * develop a new diagram of the flammability limits of diffusional cool flames affected by radiation, diffusion, and convection ; * bridge the knowledge gap between the flammability limits and extinction limits of hot flames and cool flames ; * predict and answer the question at what conditions the transition from hot flame extinction to cool flames is possible and verify the phenomena observed on the ISS flame extinguishment experiments ; * obtain new, quantitative data of extinction limit, flame temperature, and species distribution for the validation of kinetic and radiation models involving cool flames ; * extend the cool flame experiments from the original ISS droplet experiments to a more general ground-based combustion system involving radiation, diffusion, convection, and low-temperature chemistry while maintaining a one-dimensional geometry. The proposed new ground-based experiments and data analysis are complimentary to and not directly linked with the science objectives from the original ISS combustion science investigation, and will significantly advance the understanding of near-limit cool flame dynamics and contribute to the future control of engine knocking, ground and space fire safety, and combustion emissions. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | An experimentally validated dimethyl ether (DME) and methane cool flame kinetic model was developed. The results showed that the existing kinetic model significantly over-predicts the cool flame extinction limits. However, by targeting the exothermic reactions of cool flames using the cool flame extinction limits, an updated chemical kinetic model for dimethyl ether/methane mixtures is developed and validated. This study shows the value of the ozone-assisted counterflow cool flame platform in examining the key reactions contributing to the heat release rate in cool flames.<br>Diffusion warm flames of n-alkanes were successfully established. The warm flame of n-alkanes is dramatically influenced by the reactant concentrations and mixture temperatures. The primary cool flame stage is favored at lower temperatures, but the secondary warm flame stage is favored at elevated temperatures. Moreover, near extinction, the multistage warm flame collapses to a primary oxidation zone, governed by peroxy chemistry, whereas the secondary oxidation stage, governed by intermediate chemistry, is dominant near autoignition conditions. The results show that warm flames serve as a critical bridge between cool flames and hot flames.<br><br>The counterflow cool flame extinction provides a good validation target and explanation for the NASA droplet cool flame extinction limit. In addition, the results benefit the development of efficient and low mission advanced engines for transportation. |
| **Task Progress:** | In Year 4 (2019-2020), we conducted three research projects. Firstly, we used the experimental conditions of FLEX flight and conducted a comparative studies of the oxygen concentration effects on the extinction limits of cool flames. The results showed that the counterflow diffusion cool flame extinction limits are significantly affected by the oxygen concentration on the oxidizer side. However, different from the NASA FLEX data, counterflow diffusion cool flame extinction limit is more sensitive to the oxygen concentration.<br>Secondly, we conducted experimental studies of NO kinetic effects on the n-dodecane cool and warm diffusion flames. It was observed that NO addition reducing narrows cool flame extinction limits, inhibiting inhibits the transition from warm flame to cool flame, and promoting promotes the ignition transition from warm flame to hot flame. In addition, jet-stirred reactor (JSR) experiments of n-dodecane oxidation without and with NO addition were also performed. A validated low temperature n-dodecane/NOx kinetic model was developed.<br><br>Thirdly, autoignition-assisted nonpremixed cool flames of diethyl ether (DEE) are investigated under both laminar counterflow and turbulent jet flame configurations. The results show that below a critical mixture condition, the cool flame extinction limit and the low-temperature ignition limit merges at highly diluted DEE conditions, leading to an autoignition-assisted cool flame stabilization without hysteresis. By using a Co-flow Axisymmetric Reactor-Assisted Turbulent (CARAT) burner, the lift-off heights of auto-ignition assisted turbulent cool flame are quantified using formaldehyde planar laser-induced fluorescence. A correlation is proposed such that the autoignition assisted cool flame lift-off height scales with the product of the flow velocity and the square of the first-stage ignition delay time. The scaling results indicated that the correlation is valid with a chemical kinetic model which most reasonably predicted the cool flame stabilization limits of the laminar experiments. |
| **Bibliography Type:** | Description: (Last Updated: 11/26/2025) |
| **Abstracts for Journals and Proceedings** | Murakami Y, Reuter CB, Yehia OR, Ju Y. "Studies of autoignition-assisted nonpremixed cool flames." Accepted for 38th International Symposium on Combustion, Adelaide, Australia, January 24-29, 2021. (Most likely online meeting.)<br>Abstracts, in press as of May 2020. 38th International Symposium on Combustion, Adelaide, Australia, January 24-29, 2021. , May-2020 |
| **Abstracts for Journals and Proceedings** | Zhou M, Yehia OR, Reuter CB, Burger CM, Murakami Y, Ju Y. "Experimental studies of NO kinetic effects on the n-dodecane cool and warm diffusion flames." Accepted for 38th International Symposium on Combustion, Adelaide, Australia, January 24-29, 2021. (Most likely online meeting.)<br>Abstracts, in press as of May 2020. 38th International Symposium on Combustion, Adelaide, Australia, January 24-29, 2021. , May-2020 |

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| **Articles in Peer-reviewed Journals** | Ju Y, Reuter CB, Yehia OR, Farouk TI, Won SH. "Dynamics of cool flames." Progress in Energy and Combustion Science. 2019 Nov;75:100787. https://doi.org/10.1016/j.pecs.2019.100787 , Nov-2020 |